PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

ENTERPRISE SERVICE BUS

AN INTERNSHIP REPORT

Submitted by

NAVANEETH A D – 20221CAI0024

Under the guidance of,

Ms. PRIYANKI DUTTA

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING
(Artificial Intelligence &Machine Learning)

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report **"Enterprise Service Bus"** is a Bonafide work of **NAVANEETH A D (20221CAI0024),** who has successfully carried out the internship work and submitted the report for partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING (SPECIALIZATION DIVISION), ARTIFICIAL INTELLIGENCE & MACHINE LEARNING during 2025-2026.

20/4/2026

Ms. Priyanki Dutta
Assistant Professor
Internship Supervisor
PSCS (Spl. Division)
Presidency University

20/4/2026

Mr. Anandan Bellie
Assistant Professor
Program Internship Coordinator
PSCS (Spl. Division)
Presidency University

20/4/26

Dr. Geetha A
Associate Professor
School Internship Coordinator
PSCS (Spl. Division)
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20/04/2026

Dr. Zafar Ali Khan N
Professor & Head of the Department
PSCS (Spl. Division)
Presidency University

Dr. Shakkeera L
Associate Dean
PSCS (Spl. Division)
Presidency University

Sl. No	Name of the Examiner	Signature
1	Mr. VENKATESAN. D	D. Venkatesan
2.	Ms. BHAVANA. S	Bhavana

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

I am a student of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING), at Presidency University, Bengaluru, named Navaneeth A D , hereby declare that the internship work titled **"Enterprise Service Bus"** has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING (Spl.Division) (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING) during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

NAVANEETH A D USN: 20221CAI0024 Signature

PLACE: BENGALURU

DATE:20 April 2026

INTERNSHIP COMPLETION CERTIFICATE

INNOVITEGRA SOLUTIONS

CIN: U62013KA2023PTC177507
GSTIN: 29AAHCI2695D1ZC
PAN: AAHCI2695D

INTERNSHIP COMPLETION CERTIFICATE

This is to certify that Mr. **Navaneeth A D** has successfully completed an internship at our organization as a **Python Developer Intern** from **5th January 2026 to 7th April 2026**.

During this period, he worked on multiple projects, including **Enterprise Service Bus** and **ID Card OCR Reading**, both developed using Python and leveraging Artificial Intelligence and Machine Learning techniques. He contributed to tasks such as data preprocessing, model training and evaluation, and the implementation of both supervised and unsupervised learning methods.

He demonstrated a strong understanding of AI/ML fundamentals and consistently showed initiative, problem-solving abilities, and a collaborative attitude.

We appreciate his dedication and the quality of work delivered during the internship, and we wish him continued success in his academic and professional journey.

Date of Issue: 7th April 2026
Place: Bangalore

Innovitegra Solutions Private Limited Bangalore

Authorized Signatory
Director

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ACKNOWLEDGEMENTS

For completing this internship work, I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. I extend my gratitude to our beloved **Chancellor, Pro-Vice Chancellor, and Registrar** for their support and encouragement in completion of the internship.

I would like to sincerely thank my internal guide **Ms. Priyanki Dutta**, Presidency School of Computer Science and Engineering (Spl.Divison), Presidency University, for her moral support, motivation, timely guidance and encouragement provided to me during the period of internship work.

I am also thankful to **Dr. Zafar Ali Khan N**, Professor, Head of the Department, Presidency School of Computer Science and Engineering (Spl.Divison), Presidency University, for his mentorship and encouragement.

I express my cordial thanks to **Dr. Shakkeera L**, Associate Dean, Presidency School of Computer Science and Engineering (Spl.Divison), and the Management of Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my internship work.

I am grateful to **Dr. Geetha A, PSCS** (Spl.Divison), Internship Coordinator and **Mr Anandan Bellie Program** Internship Coordinator, Program Internship Coordinator, Presidency School of Computer Science and Engineering (Spl.Divison), or facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

I am also grateful to Teaching and Non-Teaching staff of Presidency School of Computer Science and Engineering (Spl.Divison), and also staff from other departments who have extended their valuable help and cooperation.

NAVANEETH A D

ABSTRACT

Modern banking and enterprise systems rely on the exchange of structured data between multiple heterogeneous backend services. These systems operate using different message formats such as JSON, XML, ISO 8583, and ISO 20022, making direct communication complex and difficult to maintain. Traditional point-to-point integrations result in tightly coupled architectures, increased transformation complexity, and reduced scalability. To address these challenges, middleware-based solutions such as Enterprise Service Bus (ESB) architectures are widely adopted to enable standardized and efficient backend communication.

This internship focuses on the design and development of a **backend-driven Enterprise Service Bus (ESB) Testing Interface** for banking message transformation and canonical processing. The system is implemented using **Python and Flask**, and exposes API-based endpoints to handle message processing workflows. It accepts requests in multiple formats, validates the incoming data, and converts it into a standardized **canonical data model**. The canonical data is then routed through a transformation pipeline and converted into the required target format using protocol-specific adapters. The backend system includes modules for inbound security validation, request-type contract enforcement, action routing, canonical mapping, and adapter execution, ensuring a modular and extensible architecture. The design emphasizes loose coupling between services, allowing independent evolution of components and easier integration of additional message formats in the future.

The developed system successfully demonstrates the ability to process and transform heterogeneous message formats through a centralized backend middleware layer. It ensures consistent data handling using the canonical model and reduces redundancy in transformation logic. The API-based design enables efficient validation and execution of transformation workflows in a controlled environment. Additionally, the system provides flexibility for extending business rules and adapting to new integration requirements without major architectural changes. Overall, the project highlights how ESB-based backend architectures improve interoperability, scalability, and maintainability in enterprise and banking systems, while providing a practical implementation of middleware concepts.